Phase 5 — Controlled End-to-End Dry Run Report

SWIFT-U5 Under-Five Mortality Prediction

Date: 2026-11-07 · Git HEAD: `9e4fdf4` · Python: 3.9 ARM64

Step 1 — Correction Verification (Source Code)

All 10 source-level checks passed. Evidence per design requirement:

#	Requirement	File / Function	Verdict
1a	`completed_5yr_followup` does not filter the cohort	`cohort.py` `filter_exposure_complete_cohort()` — only filter: `age_at_survey_months ∈ [0, 59]`	[PASS]
1b	Outcome = recorded U5 death vs no recorded death	`cohort.py` line 105: `target_u5m = (is_dead & b7_numeric < 60)`	[PASS]
1c	PSU/strata scoped to survey wave	`cohort.py`: `psu_wave = round + "_" + v021`; `strata_wave = round + "_" + v022`	[PASS]
1d	`b5`, `b7` absent from `CANDIDATE_FEATURES`	`build_features.py`: neither string appears	[PASS]
1e	Module C calibrated on held-out 2017-18 validation, not in-sample	`swift_u5.py` `fit()`: `df_val` path feeds held-out predictions to `module_c.fit_recalibration()`; fallback emits `UserWarning`	[PASS]
1f	2022 test outcomes do not enter feature selection	`select.py`: `'2022'` not present; exp03 filters to dev rounds	[PASS]
1g	Bootstrap defaults to `psu_wave` / `strata_wave`	`bootstrap.py`: `psu_col='psu_wave'`, `strata_col='strata_wave'`	[PASS]
1h	Failed fits logged explicitly	`module_b_shrink.py`: `fallback_log_` dict; `module_c_recalib.py`: `fit_failures_` dict	[PASS]
1i	OHE fitted from train categories only	`splits.py` `encode_features()`: `train_div = div_series.loc[train_idx]`; `pd.get_dummies` removed	[PASS]
1j	`PSU_COLUMN` / `STRATA_COLUMN` point to wave-scoped columns	`splits.py`: `PSU_COLUMN = 'psu_wave'`, `STRATA_COLUMN = 'strata_wave'`	[PASS]

Step 1 result: 10/10 PASS

Step 2 — Smoke Test Configuration

```
Script  : experiments/phase5_smoke_test.py
Model   : XGBoost (n_estimators=150, max_depth=4, lr=0.05)
Split   : Last temporal cut — dev 2004–2014, val 2017-18 (Module C), test 2022
Modules : A=True, B=True, C=True
Bootstrap: 10 replicates (structural check only)
Seed    : 42
Entry   : Production modules only — no simplified bypass
```

Exit code: 0 (success). Full pipeline executed without error.

Step 3 — Intermediate Output Validation

3.1 Cohort counts

Split	N	Deaths	Prevalence (uw)
Dev (2004–2014)	29,686	1,574	5.3%
Val (2017-18)	8,684	356	4.1%
Test 2022 (locked)	13,119	408	3.1%
Total	51,489	2,338	4.5%

- Six waves present: `['2004','2007','2011','2014','2017-18','2022']` ✓
- Split disjointness: dev ∩ val = ∅, dev ∩ test = ∅, val ∩ test = ∅ ✓
- `completed_5yr_followup == target_u5m` exactly (both = 2,338) ✓
- `is_exposure_complete` all-NaN (deprecated correctly) ✓
- 95.5% of the cohort (49,151 children) are alive children aged <60 months — their 5-yr survival is censored. The estimand is correctly preserved as observed death by interview date.

3.2 Leakage exclusions

Check	Result
Forbidden leakage cols (`b4`, `b19`, `m4`, `m5`, `v404`, `v201`, `h1–h10`)	[NONE]
`b5`, `b7` in parquet (expected)	Present — for audit only
`b5`, `b7` in feature matrix	✓ Absent (confirmed via `model.feature_names_`)

3.3 Survey weights

Split	w_min	w_max	w_mean	eff_N	eff_N / N
Dev	0.056	14.19	1.009	20,868	70.3%
Val	0.159	2.47	1.003	7,005	80.7%
Test	0.091	3.89	1.004	10,458	79.7%

AUDIT WARNING / RISK NOTE: Dev weights have extreme values (max=14.19). This is from the 2004 wave which has a small number of clusters with very high selection probability. This is a known DHS characteristic, not a pipeline bug. Module A clips density-ratio weights at [0.2, 5.0] to prevent destabilising influence.

3.4 Module A — weight diagnostics

Metric	Value
Density ratio classifier fitted	[TRUE]
Weight range (Module A combined)	[0.017, 180.8]
Weight mean / std	1.000 / 3.161
Weights > 5.0	3.7% of dev observations

AUDIT WARNING / RISK NOTE: Module A weight maximum = 180.8 is very extreme. The density ratio classifier is producing near-zero `P(train|x)` for some observations, causing unbounded ratios before clipping. The clip range in `module_a_weight.py` is [0.2, 5.0] for the density ratio alone — but the combined composite weight (survey × decay × density) is not globally clipped. The combined maximum of 180.8 suggests the survey weight extreme (14.19) is multiplied by a density ratio near the clip ceiling (5.0). This is a Medium-severity issue to investigate before Phase 6, but it does not block the pipeline. The effective weight of 180.8 affects a small number of observations (3.7% > 5.0). Recommendation for Phase 6: Add a final composite weight clip at e.g. 10× the median to `compute_weights()`, report ESS before and after.

Also: three `RuntimeWarning: divide by zero/overflow in matmul` from sklearn's logistic regression during density ratio fitting. These arise from near-perfect linear separation in the domain classifier — a known issue when train and target distributions differ greatly. The classifier converges (output is valid) but the warnings should be monitored.

3.5 Module B — division shrinkage

Division	lambda (shrinkage)	Weighted Deaths
barishal	0.679	84.7
chattogram	0.922	474.8
dhaka	0.947	712.5
khulna	0.839	208.8
rajshahi	0.843	215.0
rangpur	0.800	160.4
sylhet	0.833	199.9

- National intercept: −2.5603; National slope: 1.6082
- Fallback log: empty (all 7 divisions had sufficient events)
- Barishal has the lowest lambda (0.679) — smallest division, smallest death count — correctly shrunken most strongly toward national estimate ✓

3.6 Module C — calibration diagnostics

Parameter	Value
Calibration data	2017-18 held-out round (N=8,684)
Recalibration slope	0.5042
Recalibration intercept	−1.2503
Fit failures	None
In-sample calibration warning	0 (correct path used)

NOTE / IMPLEMENTATION CONTEXT: Slope = 0.50 means the base model's log-odds are over-spread: the model is too confident (over-dispersed predictions). Module C compresses the log-odds, shifting calibration toward the target round's mortality rate. This is scientifically expected — the model was trained on higher-mortality 2004–2014 rounds and is predicting into the lower-mortality 2022 environment.

3.7 Predictions on 2022 test set

Metric	Value
N test	13,119
Prob range	[0.003, 0.439]
Prob mean	0.038
Prob p95	0.075
Any NaN	[NONE]
Any out-of-[0,1]	[NONE]

3.8 Evaluation metrics (point estimates, XGBoost + All Modules, test 2022)

Discrimination

Metric	Weighted	Unweighted
AUROC	0.6256	0.6375
AUPRC	0.0727	—

Calibration

Metric	Weighted
Brier score	0.0287
Null Brier	0.0290
Brier Skill Score	0.0115
Cal slope	0.800
Cal intercept	−0.852
ECE	0.0103

Classification at Youden threshold (p=0.040)

Metric	Weighted
Accuracy	0.649
Sensitivity/Recall	0.543
Specificity	0.652
Precision	0.046
F1	0.085
Balanced accuracy	0.597
NPV	0.979
MCC	0.069

Screening utility

Capacity	Sensitivity	PPV	NNS
5%	0.180	0.109	9.2
10%	0.255	0.079	12.7
20%	0.346	0.054	18.4

All 21 required metrics present and non-NaN ✓

3.9 Bootstrap (10 replicates — structural check)

- 33 metric entries returned with CI structure ✓
- AUROC 95% CI (10-rep, illustrative): [0.603, 0.668]
- Uses `psu_wave` / `strata_wave` for resampling ✓
- Note: 10-rep CIs are extremely wide; 1,000 replicates required for publication

3.10 Decision Curve Analysis

- 60 threshold rows generated ✓
- Net benefit range: [−0.00031, 0.01992] ✓

Step 4 — Scientific Consistency vs. Frozen Research Design

Design Element	Status	Notes
Cohort: children 0–59 months at interview	[INTACT]	Filter: `0 ≤ age_at_survey_months < 60`
Six waves: 2004, 2007, 2011, 2014, 2017-18, 2022	[ALL PRESENT]	51,489 records

Design Element	Status	Notes
Outcome: recorded U5 death by interview date	[INTACT]	`b5` dead + `b7 < 60`; not confirmed survival
Follow-up completeness quantified	[NEW]	95.5% of cohort has censored 5-yr survival
T0/T1/T2 predictor tiers	[INTACT]	Defined in `FEATURE_COLUMNS`, `harmonize.py`
Feature selection: training-only	[INTACT]	`LeakageFreePreprocessor` fits train-only
2022 test set locked	[INTACT]	No tuning/calibration on test outcomes
Five forward-chaining temporal cuts	✓ All 5 present	Cuts verified
Module A: density ratio + decay	[RUNNING]	Extreme weight issue flagged (Medium)
Module B: EB division shrinkage	[RUNNING]	All 7 divisions fitted, 0 fallbacks
Module C: drift recalibration on held-out val	[FIXED] (Phase 3)	2017-18 validation round used correctly
PSU identity scoped to wave	[FIXED] (Phase 3)	360 raw overlaps → 0 wave-scoped overlaps
Bootstrap: PSU-stratified, wave-scoped	[FIXED] (Phase 3)	Defaults use `psu_wave`/`strata_wave`
Classification metrics complete	[FIXED] (Phase 3)	All 8 classification + BSS + eff_N now present

Comparison with historical benchmark numbers: The smoke test AUROC (0.626 weighted) is a valid test-set result. We cannot compare directly to the manuscript's Table 2 because:

1. The manuscript used raw `psu`/`strata` for bootstrap (now corrected)
2. Module C previously calibrated on in-sample data (now corrected)
3. The 2×2 factorial experiment (exp02) has not yet been re-run

These differences are expected given the Phase 3 corrections. Full re-run results will replace all manuscript tables.

Step 5 — Phase 5 Summary

Commands executed

```
# Step 1 verification
.venv/bin/python -c "[source inspection checks]"

# Step 2–4 smoke test
.venv/bin/python experiments/phase5_smoke_test.py

# Tests
.venv/bin/python -m unittest discover tests -v
```

Configuration

```
Model        : XGBoost (production hyperparameters)
Split        : dev=2004–2014, val=2017-18, test=2022
N_bootstrap  : 10 (structural only)
Modules A/B/C: all enabled
Seed         : 42
```

Files generated

- `outputs/reports/phase5_smoke_test_report.json` — full structured JSON output
- `experiments/phase5_smoke_test.py` — reusable smoke test script

Pipeline stages

Stage	Status
Load parquet	[SUCCESS]
Leakage check	[PASS]
PSU identity verification	[PASS]
Follow-up completeness diagnostics	[PASS]
Temporal split construction	[PASS]
Survey weight diagnostics	[PASS]
SWIFT-U5 fit (Modules A+B+C)	[SUCCESS]
Module A diagnostics	[PASS] (extreme weights flagged)
Module B diagnostics	[PASS] (0 fallbacks, 7 divisions)
Module C diagnostics	[PASS] (held-out val, no failures)
Predict on test set	[PASS] (no NaN, in [0,1])
Evaluation metrics	[ALL PRESENT]
Bootstrap (10-rep)	[PASS] (33 metrics with CIs)
DCA	[PASS] (60 threshold rows)

Stage	Status
Six-wave integrity	[PASS]
Scientific consistency checks	[8/8 PASS]

Test results

```
Ran 38 tests in 0.556s
OK  (21 original + 17 Phase 3 correction tests)
```

Open Issues Before Phase 6

#	Severity	Issue	Recommendation
A	Medium	Module A extreme composite weight (max=180.8). Survey weight (14.2) × density ratio near clip ceiling produces very high composite.	Add global composite weight clip (e.g., 99th-percentile cap) in `compute_weights()`. Report ESS before/after. Investigate 2004 high-weight cluster.
B	Low	`RuntimeWarning: divide by zero/overflow in matmul` from sklearn LR during density ratio fitting (domain classifier). Near-perfect separation.	Switch domain classifier to `max_iter=1000` or use a weaker `C=0.01`. Results are still valid but warnings are noisy.
C	Low	`encode_features()` signature change (`division_categories` param) not yet propagated to `04_run_swift_u5_ablation.py`. The ablation experiment uses `LeakageFreePreprocessor` (safe), but should be verified.	Read exp04 before Phase 6 to confirm it doesn't call `encode_features()` directly.
D	Note	Manuscript Tables 2–5 contain results from the pre-correction pipeline. All bootstrap CIs used raw PSU identity and in-sample Module C calibration.	Re-run all 5 experiments in Phase 6 to generate corrected tables.

Phase 6 Readiness Verdict

> CRITICAL DESIGN PROTOCOL: PHASE 6 READY: YES — all structural, leakage, scientific consistency, and pipeline checks passed.

Before starting Phase 6:

1. ✓ Authorize the full re-run (you'll say "run Phase 6")
2. Optionally address Issue A (Module A weight clipping) — Medium priority, can be done in Phase 6 as first step
3. Full execution order:

```
experiments/02_run_2x2_benchmark.py    (N_BOOTSTRAP=1000, all models, all 4 arms)
experiments/03_run_feature_selection.py (dev rounds only)
experiments/04_run_swift_u5_ablation.py (with df_val=2017-18)
experiments/05_run_policy_equity.py
```

4. After re-run: replace all manuscript tables and figures with corrected outputs

Stop here — Phase 6 will not begin until you authorize it.